

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0760

Poages Mill 138 kV
April 15, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0760. The Interconnection Customer proposes a 269.8 MW solar photovoltaic facility interconnecting at the Poages Mill 138 kV substation in the Virginia area. The request was received on March 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 335 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,774,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Floyds Knobs – Poages Mill 138.0 kV	Poages Mill – Meadowview 138.0 kV Line	115.7	223	100.0
Person – Riverview 345.0 kV line	Leesville – Clover 345.0 kV	107.1	1,286	24.8
Bear – Clover 345.0 kV	Person – Mecklenburg Power Sta 345.0 kV	101.7	2,871	32.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Floyds Knobs – Poages Mill 138.0 kV	Rebuild line with higher-capacity conductor	\$4,061,000	100% (\$4,061,000)
Person – Riverview 345.0 kV line	Reconductor line	\$27,654,000	7% (\$2,038,000)
Bear – Clover 345.0 kV	Reconductor line	\$42,461,000	6% (\$2,675,000)
	Total allocated to project		\$8,774,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.